VANISHING AND COMPUTATION OF PURE EXT GROUPS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file adequate.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof delivers part (3); parts (1) and (2) are the same vanishing statements that the chapter uses earlier and are not separately argued in the span. This is recorded here for completeness.

Editorial note. The counted spans carry navigation stubs for the references leaving them; no bibliography citation occurs in the spans.

Editorial note. The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

Allowed standard prerequisites (editorial).

- The two spectral sequences of a double complex and their degeneration. Homology, Section homology-section-double-complex, invoked at proof lines 2933-2936. Turns the row and column computations into the identification of the cohomology.

Source reference. Original section 012X: sources/stacks-chow/tags.txt, source lines 1415–1415. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let A be a ring.*

(1) *$Pext^i_A(M, N) = 0$ for $i > 0$ whenever N is pure injective,*
(2) *$Pext^i_A(M, N) = 0$ for $i > 0$ whenever M is pure projective, in particular if M is an A-module of finite presentation,*
(3) *$Pext^i_A(M, N)$ is also the ith cohomology module of the complex $\mathrm{Hom}_A(P_\bullet, N)$ where $P_\bullet$ is a pure projective resolution of M.*

Proof. To see (3) consider the double complex

$$A^{\bullet,\bullet} = \mathrm{Hom}_A(P_\bullet, I^\bullet)$$

Each of its rows is exact except in degree 0 where its cohomology is $\mathrm{Hom}_A(M, I^q)$. Each of its columns is exact except in degree 0 where its cohomology is $\mathrm{Hom}_A(P_p, N)$. Hence the two spectral sequences associated to this complex in Homology, Section 012X degenerate, giving the equality. □

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